

RetinalAI: Hackathon & Pitch Guide

Classification: Presentation Playbook

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Purpose: Step-by-step guide for presenting RetinalAI at hackathons, investor pitches, and demo days

1. Before the Event: Preparation Checklist

Technical Setup (Do 1 Hour Before)

```
# 1. Start the full stack
docker compose up -d

# 2. Verify backend health
curl http://localhost:8088/health

# 3. Verify frontend
open http://localhost:3330

# 4. Pre-load a test prediction to warm the model cache
curl -X POST http://localhost:8088/predict -F "file=@test_images/sample_fundus.jpg"

# 5. Clear browser cache and open fresh session
# 6. Set display to 1920x1080, 100% zoom
# 7. Disable all OS notifications
# 8. Keep a mobile hotspot as network backup
```

Demo Images Ready

Image	Expected Result	Purpose
Healthy fundus	Normal / No disease	Shows the system handles negatives correctly
Diabetic retinopathy (severe)	DR + related conditions, URGENT referral	The "wow" moment — multi-label + clinical reasoning
Glaucoma suspect	ODC/ODP, ROUTINE referral	Shows referral priority variation
Mixed pathology	Multiple conditions, knowledge graph adjustments visible	Demonstrates clinical knowledge graph reasoning

Backup Plan

- **If the live demo fails:** Have 2-minute screen recording of a successful demo ready
- **If network fails:** Backend and frontend both run locally in Docker
- **If projector/screen fails:** Have key screenshots on your phone for a verbal walkthrough

2. The 3-Minute Hackathon Pitch

Context: Fast-paced, technical judges, 20+ teams presenting, attention spans are short.

Goal: Make them remember you. Problem-demo-impact.

Structure

```
[0:00 - 0:20] THE HOOK
[0:20 - 0:50] THE PROBLEM
[0:50 - 2:10] LIVE DEMO
[2:10 - 2:40] TECH DIFFERENTIATORS
[2:40 - 3:00] IMPACT & CLOSE
```

Script

THE HOOK (0:00 - 0:20)

"There are 285 million people with visual impairment worldwide. 80% of blindness is preventable — if caught early. In Sub-Saharan Africa, there's one ophthalmologist for every million people. We built RetinalAI — a system that screens for 45 eye diseases from a single photograph in under 200 milliseconds, with clinical reasoning a doctor can trust."

[RetinalAI dashboard is already loaded on screen]

THE PROBLEM (0:20 - 0:50)

"Existing AI solutions detect 1-3 diseases, cost \$25-40 per test, and give you a probability with zero explanation. That means a patient needs multiple tests, a clinic needs a budget, and a doctor needs to trust a black box. In places like Uganda, patients wait months for a single screening. By the time they're seen, the damage is irreversible."

LIVE DEMO (0:50 - 2:10)

"Let me show you what we built."

[Navigate to Screening page]

"I'm uploading a retinal fundus image from the RFMiD clinical dataset."

[Drag-and-drop image → Click 'Analyze Image']

"The image first passes through our 3-layer fundus validation gate — checking that this is actually a retinal photograph. Then our SceneGraphTransformer — a Graph Neural Network with uncertainty calibration — classifies all 45 diseases simultaneously."

[Results appear]

"In 205 milliseconds: 11 conditions detected, URGENT referral priority. Look at this — the system didn't just predict diseases in isolation. Our Clinical Knowledge Graph — encoding 144 disease relationships from peer-reviewed literature — boosted the probability of Cystoid Macular Edema because it co-occurs with Diabetic Retinopathy. This is clinical reasoning, not just pattern matching."

[Click GradCAM tab]

"And we don't ask doctors to trust a black box. GradCAM shows exactly which retinal regions drove each prediction. We also support LIME, SHAP, and Integrated Gradients — four explainability methods in production."

TECH DIFFERENTIATORS (2:10 - 2:40)

"Under the hood: four custom GNN architectures evaluated in rigorous cross-validation. Full MLOps pipeline — data validation, drift detection, model governance, EU AI Act compliance. Exported to ONNX and TorchScript. Runs on CPU for resource-constrained settings. 68 automated tests. This is not a notebook — it's a production system."

IMPACT & CLOSE (2:40 - 3:00)

"One photograph. 45 diseases. 200 milliseconds. At 50 cents a scan — 50x cheaper than any competitor — we can screen every diabetic patient in Uganda for less than the cost of a text message. RetinalAI. Early detection saves sight."

[Hold for applause / transition to Q&A]

3. The 5-Minute Investor Pitch

Context: Investor audience, looking for market size, defensibility, and team.

Goal: Get the follow-up meeting.

Structure

[0:00 - 0:30]	HOOK + MARKET PROBLEM
[0:30 - 1:30]	LIVE DEMO (condensed)
[1:30 - 2:30]	CLINICAL KNOWLEDGE GRAPH (the moat)
[2:30 - 3:30]	MARKET & BUSINESS MODEL
[3:30 - 4:30]	TRACTION & REGULATORY
[4:30 - 5:00]	THE ASK

Script

HOOK + MARKET PROBLEM (0:00 - 0:30)

"2.2 billion people worldwide have vision impairment. There are only 232,000 ophthalmologists to serve them. The ophthalmic AI market is \$5.4 billion and growing at 31% annually. Every existing solution screens for 1-3 diseases at \$25-40 per test. We screen for 45 diseases at 50 cents."

LIVE DEMO (0:30 - 1:30)

[Same demo flow as hackathon, slightly faster — upload, results, GradCAM. 60 seconds.]

"One scan. 45 diseases. 200 milliseconds. Explainable results with referral prioritization."

CLINICAL KNOWLEDGE GRAPH — THE MOAT (1:30 - 2:30)

"Here's why this is hard to replicate. Our Clinical Knowledge Graph encodes 144 disease relationships from peer-reviewed ophthalmology literature — co-occurrence patterns, severity hierarchies, and population-specific prevalence data calibrated for Uganda's disease epidemiology."

"When our model detects Diabetic Retinopathy with high confidence, the knowledge graph automatically adjusts related condition probabilities — just like a specialist would reason through a differential diagnosis. This isn't a model weight that any team with GPUs can reproduce — it's structured clinical knowledge that took years of ophthalmological expertise."

"Combined with our four GNN architectures and four explainability methods, this creates compounding technical defensibility."

MARKET & BUSINESS MODEL (2:30 - 3:30)

"Our go-to-market follows the classic 'start where you're the only option' playbook."

"Phase 1 — East Africa. One ophthalmologist per million people. No AI competition. We're already calibrated for local disease prevalence. We partner with Ministries of Health at 30-50 cents per scan."

"Phase 2 — UK and EU. CE Mark pathway is ready, and we're already EU AI Act compliant — most competitors will need to retrofit. NHS AI Diagnostic Fund is actively seeking solutions like ours."

"Phase 3 — US. FDA De Novo submission. No predicate exists for 45-disease screening — we become the predicate. That's a regulatory moat."

"Revenue model: per-scan fees for government programs, SaaS subscriptions for clinics, enterprise licenses for hospital networks, OEM royalties for camera manufacturers embedding our API. Blended gross margin: 76%."

"Year 1 target: \$1.76M ARR. Year 3: \$25.5M. Break-even at month 18."

TRACTION & REGULATORY (3:30 - 4:30)

"Where we are today: working product with 45-disease classification, full MLOps pipeline, 68 automated tests, EU AI Act compliance embedded from day one. Four trained model architectures with published performance benchmarks."

"Regulatory pathway: CE Mark submission Q4 2026, FDA Pre-Sub meeting Q3 2026, East African registration Q1 2027. We've already engaged with Mulago National Referral Hospital in Kampala for our first clinical validation site."

"IP: First provisional patent filing Q3 2026 — clinical knowledge graph-augmented GNN for multi-label medical image classification. Three additional patents in the pipeline."

THE ASK (4:30 - 5:00)

"We're raising a \$2-4 million Seed round to execute clinical validation, CE Mark submission, and first commercial pilots."

"Use of funds: 40% clinical validation and regulatory, 30% engineering and infrastructure, 20% first three hires — VP Regulatory, Clinical Director, ML Engineer — and 10% operating reserve."

"We'll hit three milestones before the next raise: CE Mark granted, retrospective study published, and first paying customers in East Africa and the UK."

"We're making AI screening possible for the 4 billion people who can't access an eye specialist. Let's talk."

4. The 10-Minute Demo Day Presentation

Context: Mix of investors, partners, press, and peers. More time for depth.

Goal: Full understanding of product, technology, market, and team.

Structure

```
[0:00 - 1:00] PROBLEM NARRATIVE
[1:00 - 3:00] FULL LIVE DEMO (with clinical reasoning deep-dive)
[3:00 - 4:00] EXPLAINABILITY SHOWCASE
[4:00 - 5:30] ARCHITECTURE & MLOPS
[5:30 - 7:00] MARKET & COMPETITIVE POSITIONING
[7:00 - 8:30] BUSINESS MODEL & FINANCIALS
[8:30 - 9:30] REGULATORY & TRACTION
[9:30 - 10:00] VISION & CLOSE
```

Key Additions (vs. 5-Minute Pitch)

Problem Narrative — Tell the story of a specific patient:

"Grace is a 54-year-old diabetic woman in Gulu, Northern Uganda. The nearest ophthalmologist is in Kampala — 340 kilometers away. She's been losing peripheral vision for two years, but a screening would cost more than her monthly income. By the time she saves enough to travel, she has advanced glaucoma with irreversible vision loss. This story repeats 80 million times a year across Africa."

Architecture Deep-Dive — Show the technical sophistication:

"We evaluated four custom graph neural network architectures in rigorous 3-fold cross-validation: ViGNN with disease prototypes, GraphCLIP with sparse attention, VisualLanguageGNN with cross-modal fusion, and our winning SceneGraphTransformer with 3 ensemble branches and uncertainty calibration. All share a pretrained Vision Transformer backbone and our Clinical Knowledge Graph."

MLOps Pipeline — Emphasize production readiness:

"This isn't a Jupyter notebook demo. Data ingestion with 7 automated quality checks. Training with differential learning rates, EMA, mixed precision. Model comparison with weighted scoring. Export to ONNX and TorchScript with quantization. Monitoring with drift detection and SLA compliance. Governance with model cards, fairness evaluation, and a human-in-the-loop review queue. Every prediction is logged. Every model version is tracked."

5. Q&A Preparation: Top 20 Questions & Answers

Technical

Q: How does this compare to Google's ARDA system?

"Google ARDA screens for one disease — diabetic retinopathy — and isn't commercially available. It's a research project, not a product. We screen for 45 diseases, we're deploying to production, and our clinical knowledge graph adds reasoning that pure deep learning approaches lack."

Q: What's your training data? How do you handle class imbalance?

"We train on the RFMiD dataset — 3,200 expert-annotated fundus images across 45 disease labels. Class imbalance is severe — some diseases appear in <1% of images. We use asymmetric loss functions, focal loss weighting, and our clinical knowledge graph acts as an implicit regularizer by constraining predictions to clinically plausible combinations."

Q: 45 diseases sounds like you're overfitting. What's your real-world accuracy?

"Valid concern. In 3-fold cross-validation, our SceneGraphTransformer achieves the highest F1 score with 80% sample accuracy. Critically, we're not claiming 90%+ accuracy on all 45 diseases. We're launching with the top 10 clinically validated diseases and expanding as prospective studies confirm performance. The 45-disease architecture is the platform — the clinical claims follow the evidence."

Q: Can this run without internet?

"Yes. The entire system runs in Docker. CPU inference is supported. We've tested deployment on modest hardware. For LMIC settings, we plan edge deployment — the model runs locally, results sync when connectivity is available."

Q: How is this different from just fine-tuning a Vision Transformer?

"A fine-tuned ViT gives you probabilities. Our system gives you clinical reasoning. The Knowledge Graph encodes that Diabetic Retinopathy co-occurs with Macular Edema, that Glaucoma has severity stages, that Central RVO is an emergency. A ViT doesn't know any of this. Our GNN architectures explicitly model disease relationships as graph structures."

Clinical / Regulatory

Q: Is this FDA cleared?

"Not yet — we're pursuing a De Novo pathway because no predicate exists for 45-disease multi-label screening. FDA Pre-Sub meeting targeted Q3 2026, submission by Q1 2027. CE Mark is on a faster timeline — Q4 2026. We're launching in East Africa first, where registration is faster and the need is most acute."

Q: What about liability if the AI misses something?

"RetinalAI is positioned as a clinical decision support system — a triage and screening aid — not an autonomous diagnostic. The ophthalmologist makes the final diagnosis. Our referral priority ranking (Emergency / Urgent / Routine) is designed to ensure high-risk cases are escalated to specialists, not replaced by AI. Our intended use statement and labeling will be aligned with FDA/CE expectations for this positioning."

Q: How do you handle different cameras and image quality?

"Our 3-layer fundus validation gate checks image structure, statistical properties, and confirms it's a retinal photograph before prediction. If image quality is insufficient, the system rejects the image and asks for re-capture. We've tested with images from multiple camera types in the RFMiD dataset."

Q: What's your plan for post-market surveillance?

"Built in from day one. Every prediction is logged with timestamps, confidence scores, and model versions. Drift detection monitors for distribution shifts. The governance module includes fairness evaluation across demographic groups. The audit trail is SHA-256 chained for regulatory integrity."

Business / Market

Q: Why not just sell to the US first? That's where the money is.

"The US is the most expensive market to enter — FDA clearance costs \$200-350K, the sales cycle for hospitals is 6-12 months, and you need clinical evidence that we don't have yet. East Africa is where we build clinical evidence with real patients, refine the product, and generate our first revenue — all at 1/10th the cost. By the time we enter the US, we have prospective clinical data, CE Mark approval, and paying customers. That's a much stronger position for FDA and for enterprise sales."

Q: How do you compete with incumbents like IDx-DR?

"We don't compete — we complement. IDx-DR screens for one disease at \$40/test. We screen for 45 at \$0.50. They serve US retinal specialists with Topcon cameras. We serve East African primary care clinics with any fundus camera. The overlap is minimal today. By the time it isn't, we'll have regulatory approvals, clinical evidence, and OEM partnerships."

Q: What if Google or a big tech company enters this space seriously?

"Google has been in ophthalmic AI since 2016 and hasn't launched a commercial product. Building the model is 20% of the problem — regulatory clearance, clinical validation, hospital integration, and clinical knowledge graph calibration for specific populations is the other 80%. Our LMIC-first strategy gives us clinical data and real-world deployment experience that big tech cannot easily replicate from Mountain View."

Q: Your revenue projections assume fast clinic adoption. What's the actual sales cycle?

"For Tier 2 (specialty clinics): 2-3 months. Low price point, clear ROI, standard integration. For Tier 1 (hospital networks): 4-6 months. Requires clinical validation review, IT security assessment, procurement. For Tier 3 (government): 6-12 months. Tender-based, often requires on-premise deployment. Our Year 1 projections are conservative — 5 enterprise + 30 clinics + 2 government programs."

Q: What's your burn rate and runway?

"Post-Seed at \$3M: approximately \$150K/month. That gives us 20 months of runway — enough to reach CE Mark, first revenue, and Series A milestones. If revenue comes faster, we extend. If regulatory takes longer, we have buffer."

Impact / Vision

Q: What's the long-term vision beyond retinal screening?

"RetinalAI is a platform, not a point solution. The retina is the only place in the body where you can non-invasively photograph blood vessels and neural tissue. We're starting with eye diseases, but the same image can screen for diabetes, hypertension, cardiovascular risk, and neurological conditions. Our clinical knowledge graph architecture is modality-agnostic — it can incorporate OCT, slit-lamp images, and eventually non-ophthalmic imaging. The long-term vision is the AI-powered screening gateway for preventive healthcare."

Q: How do you measure impact beyond revenue?

"Patients screened, diseases detected, referrals generated, and — most importantly — sight-threatening conditions caught early that would otherwise have progressed to irreversible vision loss. We'll track time-from-screening-to-treatment as our north star clinical outcome metric."

6. Stage Presence & Delivery Tips

Voice & Pacing

- **Slow down for numbers:** "forty-five diseases... two hundred milliseconds... fifty cents per scan" — let them land
- **Speed up for technical details:** Architecture names, MLOps features — confidence, not lecture
- **Pause after the hook:** Let the problem statement sit for 2 seconds before offering the solution
- **End strong:** The last sentence should be a declarative statement, not a question. Drop your pitch, look at the audience.

Body Language

- **Stand to the side of the screen,** not in front of it — the demo is the star
- **Point at specific UI elements** when you mention them — guide their eyes
- **Make eye contact with judges/investors** during the problem and close sections
- **Hands visible** — no pockets, no behind-the-back. Open gestures build trust

Demo Execution

- **Pre-position the image** in a visible folder on the desktop — no hunting through file systems
- **Use drag-and-drop,** not file picker — it looks more natural and is faster

- **Click deliberately** — rushed clicks make people nervous
- **If something loads slowly**: narrate what's happening ("the model is running inference across all 45 disease classes") rather than standing in silence
- **If the demo breaks**: "Let me show you what this looks like" and switch to the backup recording without apologizing excessively. One "technical difficulties" acknowledgment is fine. Then move on.

Common Mistakes to Avoid

1. **Don't read from slides or notes** — know your pitch cold
2. **Don't apologize for what the product doesn't do yet** — focus on what it does
3. **Don't use jargon without context** — "GradCAM" means nothing to a business judge. Say "visual explainability heatmaps" first, then name the method
4. **Don't rush the demo** — the live demo is your strongest asset. Give it time
5. **Don't end with "any questions?"** — end with your impact statement. Questions will come
6. **Don't say "as you can see"** — instead, point and describe what they're looking at

7. Hackathon Judging Criteria Alignment

Most hackathons judge on these criteria. Here's how to emphasize each:

Criteria	How RetinalAI Scores	What to Emphasize
Innovation	45-disease multi-label with clinical knowledge graph reasoning	"No existing system does this — the knowledge graph is our innovation"
Technical Complexity	4 GNN architectures, full MLOps, ONNX export, 68 tests	Show the architecture briefly, mention cross-validation rigor
Impact / Usefulness	285M visually impaired, LMIC specialist shortage	Lead with the problem. Make it personal.
Completeness	Full-stack: training, evaluation, backend, frontend, deployment	"This is not a notebook — it's a production system"
Presentation Quality	Live demo with real predictions	The demo IS the presentation. Make it flawless.
Scalability	Docker, ONNX, CPU support, \$0.02/scan compute cost	Mention briefly — "runs on CPU, deployed via Docker, sub-\$1 per scan"

8. Pitch Deck Slide Outline (If Slides Required)

For events requiring slides alongside the demo:

Slide	Content	Time
1. Title	RetinalAI logo + tagline: "45 diseases. 200ms. 50 cents."	5s
2. Problem	Stats: 285M impaired, 1 ophthalmologist per 1M in SSA, 80% preventable	20s
3. Solution	One-line: AI screening with clinical reasoning + explainability	10s
4. LIVE DEMO	Switch to browser — this is the core of the pitch	60-120s
5. How It Works	Architecture diagram: Image → Validation → GNN → Knowledge Graph → Results	15s
6. Clinical Knowledge Graph	Before/after probabilities, disease relationships visual	20s
7. Competitive Landscape	2x2 matrix: # diseases vs. price per scan (we're top-left)	15s
8. Market	\$5.4B TAM, 31% CAGR, go-to-market phases	20s
9. Business Model	Tier table: per-scan, SaaS, enterprise, OEM	15s
10. Traction & Roadmap	CE Mark Q4 2026, FDA Q3 2027, first revenue Q1 2027	15s
11. Team	Founder backgrounds + key hires planned	10s
12. The Ask	"\$2-4M Seed to reach CE Mark + first revenue"	15s
13. Close	"One photograph. 45 diseases. 200 milliseconds. Early detection saves sight."	10s

9. Audience-Specific Adaptations

For Technical Judges (Hackathon)

- Lead with the architecture: "Four custom GNN architectures with a shared Clinical Knowledge Graph"
- Show code quality: mention 68 tests, type safety, CI/CD
- Emphasize the MLOps pipeline: "This is production-grade, not research-grade"
- Use precise numbers: "144 relationships", "3-fold cross-validation", "sub-200ms p99 latency"

For Business Judges (Hackathon)

- Lead with market size: "\$5.4 billion ophthalmic AI market"
- Emphasize unit economics: "\$0.02 compute cost, \$0.50 price, 76% gross margin"
- Show the competitive moat: "45 diseases vs. competitors' 1-3"
- Mention regulatory strategy: "EU AI Act compliant from day one"

For Investors (Pitch)

- Lead with the problem scale and urgency
- Focus on defensibility: knowledge graph + regulatory moat + LMIC-first data advantage
- Give clear financials: "\$1.76M Year 1 ARR, break-even month 18"
- End with a specific ask and specific milestones

For Clinical Audience (Conference/Hospital Demo)

- Lead with patient outcome: "Time-from-screening-to-treatment reduction"
- Show the referral priority system in detail
- Deep-dive into explainability: "You can verify exactly why the AI made each prediction"
- Emphasize the human-in-the-loop: "The AI triages; the doctor decides"
- Avoid business jargon — speak in clinical terms

For Government / Public Health Buyers

- Lead with population impact: "Screen every diabetic citizen for under \$0.50"
 - Emphasize offline capability and Docker deployment
 - Show the governance module: audit trail, model cards, fairness evaluation
 - Mention data sovereignty: "On-premise deployment, your data stays in your country"
 - Speak to WHO screening guidelines and SDG targets
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10. Post-Pitch Follow-Up Template

For Investors (Send Within 24 Hours)

Subject: RetinalAI – Follow-up from [Event Name]

Hi [Name],

Thank you for your time at [Event]. As discussed, RetinalAI screens for 45 retinal diseases in <200ms at \$0.50/scan with built-in clinical reasoning and explainability.

Key links:

- Live demo: [URL]
- GitHub: [URL]
- Technical documentation: [URL]

We're raising a \$2-4M Seed round. I'd welcome 30 minutes to walk through our clinical validation plan and regulatory pathway in detail.

Available [suggest 3 specific times].

Best,
[Your name]

For Potential Partners (Camera OEMs, Hospitals)

Subject: RetinalAI – AI screening integration for [Partner's Product/Hospital]

Hi [Name],

Great meeting you at [Event]. RetinalAI's API can integrate directly with [their camera/EHR system] to add 45-disease AI screening with clinical reasoning and referral prioritization.

Integration: REST API, Docker deployment, ONNX/TorchScript models.

Compliance: EU AI Act ready, CE Mark pathway Q4 2026.

Cost: Sub-\$1/scan at volume.

I'd love to explore a pilot program. Would [suggest time] work for a technical discussion?

Best,
[Your name]

This guide is a living document. Update after every pitch with lessons learned. What resonated? What got blank stares? What questions came up that you didn't anticipate? The pitch gets better with every iteration.